Dayananda Sagar University

First Year BTech Semester-I Batch-2024-2025

Course : Linear Algebra and Differential Equations (24EN1101)

Assignment-II

	Questions
1	**Find** the basis and dimension of the null and column spaces of the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$
2	**Show** that $\{u_1, u_2, u_3\}$ is an orthogonal basis for R^3. Then express $\begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}$ as a linear combination of the vectors $u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ and $u_3 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$.
3	Let $u_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 6 \\ -8 \\ -2 \\ -4 \end{bmatrix}$, $u_3 = \begin{bmatrix} 6 \\ 3 \\ 6 \\ -3 \end{bmatrix}$. Let W= span$\{u_1, u_2, u_3\}$. **Construct** an orthonormal basis for W using Gram-Schmidt Orthogonalization process.
4	**Show** that 10 is an eigenvalue of the matrix $\begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}$ and find the corresponding eigenvectors.
5	**Find** a matrix P which diagonalize the matrix $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$. Verify $P^{-1}AP = D$, where D is the diagonal matrix.
6	Two of the three eigen values of a 3×3matrix A are -1, 2. If the determinant of the matrix is 4, **find** the third eigen value of the matrix A.
7	**Find** the general solution of the differential equation $y'' - y' - 6y = 0;\ y(0) = 10,\ y'(0) = 1$
8	**Find** the general solution of $\frac{d^2y}{dx^2} - 25y = e^{-6x} + 3\sin 2x$ using method of undetermined coefficient.
9	**Solve** by the variation of parameters $\frac{d^2y}{dx^2} + 9y = \sec 3x$
10	**Find** the general solution of $x^2y'' + 3xy' + y = 0$